Limits

ANSWER KEY

Section 1: Algebraic Limits & Indeterminate Forms

1 Evaluate the limit: $\lim_{x \to 4} \frac{x^2 - 16}{x - 4}$.

a. 4

b. 8 ✓

c. 16

d. Does not exist

Factor numerator: $\lim_{x \to 4} \frac{(x-4)(x+4)}{x-4} = \lim_{x \to 4} (x + 4) = 4 + 4 = 8$.

2 Evaluate the limit: $\lim_{x \to 0} \frac{\sqrt{1+x} - 1}{x}$.

a. $\frac{1}{2}$ ✓

b. 1

c. 2

d. 0

Multiply numerator and denominator by conjugate $(\sqrt{1+x} + 1)$: $\frac{(1+x)-1}{x(\sqrt{1+x}+1)} = \frac{1}{\sqrt{1+x}+1} \to \frac{1}{2}$ as $x \to 0$.

3 Evaluate: $\lim_{x \to 2} \frac{\frac{1}{x} - \frac{1}{2}}{x - 2}$.

a. $-\frac{1}{4}$ ✓

b. $\frac{1}{4}$

c. $-\frac{1}{2}$

d. $\frac{1}{2}$

Combine numerator fractions: $\frac{2-x}{2x(x-2)} = \frac{-(x-2)}{2x(x-2)} = -\frac{1}{2x}$. As $x \to 2$, this equals $-\frac{1}{4}$.

Section 2: One-Sided Limits & Continuity

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4 Evaluate the one-sided limit: $\lim_{x \to 3^+} \frac{|x-3|}{x-3}$.

a. 1 ✓

b. -1

c. 0

d. Does not exist

For $x > 3$, $x - 3 > 0$, so $|x-3| = x-3$. The quotient is $\frac{x-3}{x-3} = 1$.

5 For what value of k is $f(x) = kx + 1$ for $x \leq 2$, and $f(x) = x^2 - k$ for $x > 2$ continuous at $x = 2$?

a. 1 ✓

b. $\frac{3}{2}$

c. 2

d. 3

Set left limit equal to right limit: $2k + 1 = 2^2 - k \Rightarrow 2k + 1 = 4 - k \Rightarrow 3k = 3 \Rightarrow k = 1$.

6 What is $\lim_{x \to 0^-} \frac{1}{x}$?

a. $-\infty$ ✓

b. ∞

c. 0

d. Does not exist

As x approaches 0 through negative values, $\frac{1}{x}$ decreases without bound toward $-\infty$.

Section 3: Limits at Infinity & Asymptotes

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7 Evaluate the limit at infinity: $\lim_{x\to\infty} \frac{5x^3 - 2x + 7}{3x^3 + 4x^2 - 1}$.

a. $\frac{5}{3}$ ✓

b. 0

c. ∞

d. $\frac{7}{-1}$

Degrees of numerator and denominator are equal (3). The limit is the ratio of leading coefficients: $\frac{5}{3}$.

8 Evaluate: $\lim_{x\to\infty} \frac{2x + 1}{\sqrt{9x^2 + 4}}$.

a. $\frac{2}{3}$ ✓

b. $\frac{2}{9}$

c. 0

d. ∞

Divide by x: $\lim_{x\to\infty} \frac{2 + 1/x}{\sqrt{9 + 4/x^2}} = \frac{2}{\sqrt{9}} = \frac{2}{3}$.

9 Evaluate: $\lim_{x\to\infty} (\sqrt{x^2 + 4x} - x)$.

a. 2 ✓

b. 4

c. 0

d. ∞

Multiply and divide by conjugate: $\frac{(x^2 + 4x) - x^2}{\sqrt{x^2 + 4x} + x} = \frac{4x}{\sqrt{x^2 + 4x} + x}$. Divide by x: $\frac{4}{\sqrt{1 + 4/x} + 1} = \frac{4}{2} = 2$.

Section 4: Trigonometric Limits & Definition of Derivative

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10 Evaluate the special trigonometric limit: $\lim_{x \to 0} \frac{\sin(5x)}{2x}$.

a. $\frac{5}{2}$ ✓

b. $\frac{2}{5}$

c. 1

d. 0

Rewrite as $\frac{5}{2}\lim_{x \to 0} \frac{\sin(5x)}{5x} = \frac{5}{2}(1) = \frac{5}{2}$.

11 Evaluate: $\lim_{x \to 0} \frac{1 - \cos x}{x^2}$.

a. $\frac{1}{2}$ ✓

b. 1

c. 0

d. Does not exist

Multiply by $(1 + \cos x)$: $\frac{\sin^2 x}{x^2(1 + \cos x)} = (\frac{\sin x}{x})^2 \frac{1}{1 + \cos x} \to (1)^2 \frac{1}{2} = \frac{1}{2}$.

12 The limit $\lim_{h \to 0} \frac{(2 + h)^3 - 8}{h}$ represents the derivative of which function at which point?

a. $f(x) = x^3$ **at** $x = 2$ ✓

b. $f(x) = x^3$ at $x = 8$

c. $f(x) = 3x$ at $x = 2$

d. $f(x) = (x + 2)^3$ at $x = 0$

By definition of derivative $f'(a) = \lim_{h \to 0} \frac{f(a + h) - f(a)}{h}$, this matches $f(x) = x^3$ evaluated at $a = 2$.

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